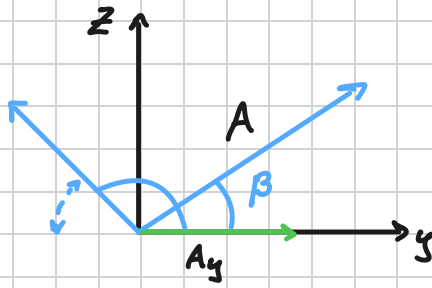
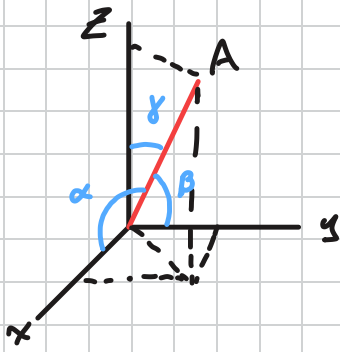


Weeks 4



$$\cos \alpha = \frac{A_x}{A}$$

$$\cos \beta = \frac{A_y}{A}$$

$$\cos \gamma = \frac{A_z}{A}$$

$$\boxed{A = A_x i + A_y j + A_z k}$$

$$\begin{array}{ccccccc}
 \uparrow & & & & & & \\
 \text{Unit vector} & = & \frac{A}{A} & = & \frac{A_x}{A} i & + & \frac{A_y}{A} j & + & \frac{A_z}{A} k \\
 & & & & \downarrow & & \downarrow & & \downarrow \\
 & & & & \cos \alpha & & \cos \beta & & \cos \gamma
 \end{array}$$

$$u_A = \frac{A_x}{A}i + \frac{A_y}{A}j + \frac{A_z}{A}k$$

$$u_A = \cos \alpha i + \cos \beta j + \cos \gamma k$$

$$u_A = \sqrt{\cos^2 \alpha i + \cos^2 \beta j + \cos^2 \gamma k}$$

$$(1.0)^2 = \left(\sqrt{\cos^2 \alpha i + \cos^2 \beta j + \cos^2 \gamma k} \right)^2$$

$$1 = \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$$

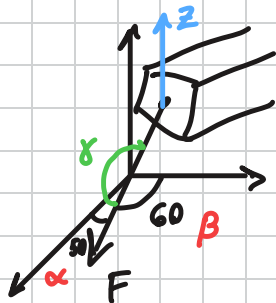
Important

Plug in to check

We can use two angles to find the third

E_x

$$F = 450 \text{ lb}$$



$$\vec{F} = \{F_x \hat{i} + F_y \hat{j} + F_z \hat{k}\}$$

$$\alpha = 50^\circ \quad \beta = 60^\circ \quad \gamma =$$

Steps

$$\textcircled{3} \quad \vec{F} = \text{magnitude} \cdot \text{direction}$$

$$\vec{F} = 450 \text{ lb} \cdot \vec{U}_F$$

$$\textcircled{2} \quad \vec{U}_F = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$$

$$\textcircled{1} \quad 1.0 = \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$$

Solve

$$1.0 = \cos^2 50^\circ + \cos^2 60^\circ + \cos^2 \gamma$$

$$\cos^2 \gamma = 1.0 - \cos^2 50^\circ + \cos^2 60^\circ$$

$$\cos \gamma = \pm 0.584$$

$$\cos \gamma = -0.584$$

$$\gamma \approx 126^\circ$$

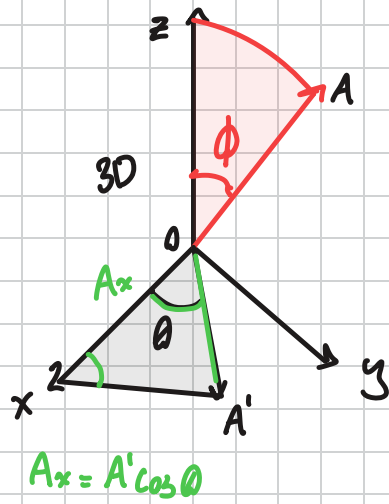
solve for γ

$$\vec{F} = 45^\circ \vec{u}_F$$

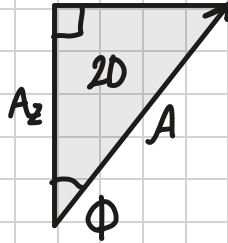
$$= 45 \{ \cos 50^\circ \hat{i} + \cos 60^\circ \hat{j} + \cos 125.48^\circ \hat{k} \}$$

$$= \left\{ \underbrace{249}_{F_x} \hat{i} + \underbrace{225}_{F_y} \hat{j} - \underbrace{261}_{F_z} \hat{k} \right\} \text{ Lb}$$

We see each directional Force
for each axis



A projected down
to show A'



$$A_z = A \cos \phi$$

in this case

$$A_z = A \cos \phi$$

$$A' = A \sin \phi$$

$$A_x = A' \cos \theta \quad A_y = A' \sin \theta$$

$$A = A \sin \phi \cos \theta \, i + A \sin \phi \sin \theta \, j + A \cos \phi \, k$$

$$A_x = A \sin \phi \cos \theta \quad A_y = A \sin \phi \sin \theta$$

θ can move in different equations

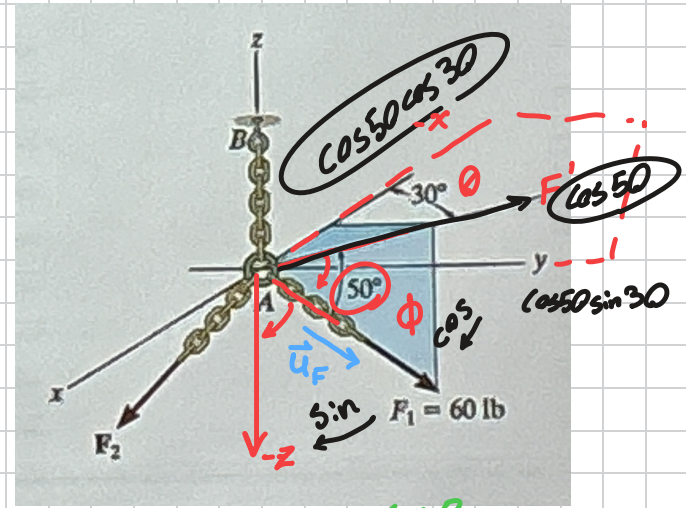
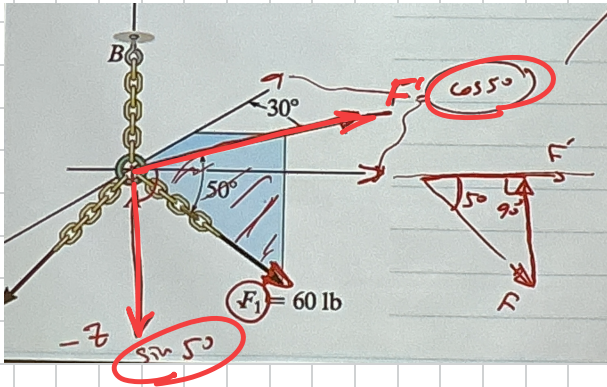
Must understand
Not memorize

α, β, γ of force F_i

$$\vec{u}_F = (\phi, \theta) \rightarrow \textcircled{1}$$

$$\vec{u}_F = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k} \rightarrow \textcircled{2}$$

$$\textcircled{1} = \textcircled{2}$$



$$\vec{u}_F = \left\{ \underbrace{-\cos 50 \cos 30}_{\cos \alpha} \hat{i} + \underbrace{\cos 50 \sin 30}_{\cos \beta} \hat{j} - \underbrace{\sin 50}_{\cos \gamma} \hat{k} \right\}$$

basically project angle and use $\cos$ or $\sin$ and that's your $\vec{u}$ vector

Using triangles we know if x, y, z is positive or negative

$$\cos \alpha = -\cos 50 \cos 30$$

$$= -0.556$$

$$\alpha = 124^\circ$$

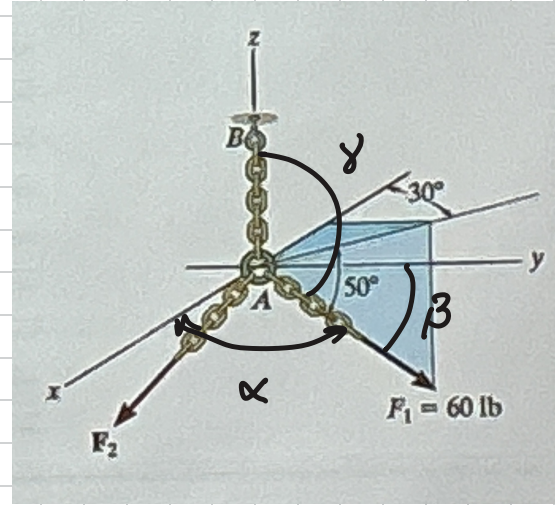
$$\cos \beta = \cos 50 \sin 30$$

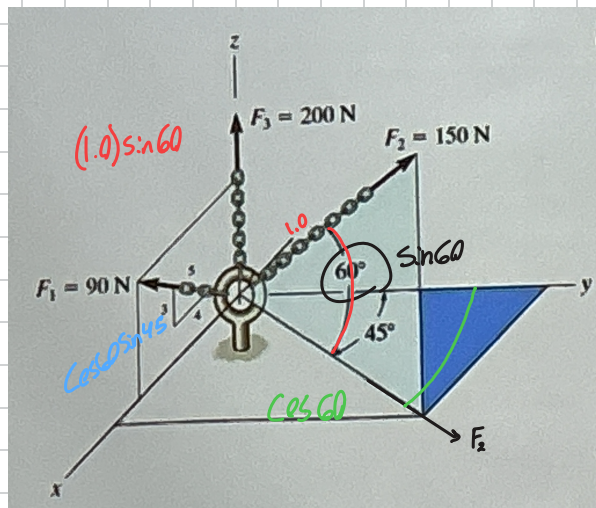
$$= 0.321$$

$$\beta = 71.3^\circ$$

$$\cos \gamma = -\sin 50$$

$$\gamma = 140^\circ$$



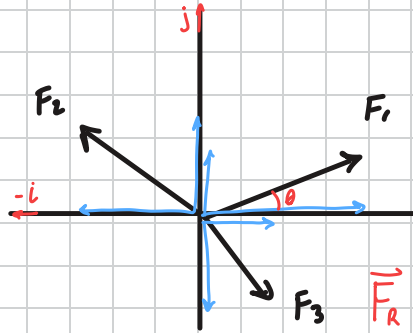


$$\vec{F}_2 = 150 \quad \vec{u}_F$$

$$u_F = \{ \quad + \sin 60 \}$$

$$\vec{u} = \{ \cos 60 \sin 45 \hat{i} + \cos 60 \cos 45 \hat{j} + \sin 60 \hat{k} \}$$

Vector addition



$$\vec{F}_1 = F_{1x}\vec{i} + F_{1y}\vec{j}$$

$$\vec{F}_2 = -F_{2x}\vec{i} + F_{2y}\vec{j}$$

$$\vec{F}_3 = F_{3x}\vec{i} - F_{3y}\vec{j}$$

$$\vec{F}_R = F_{Rx}\vec{i}$$